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CLK2 inhibition ameliorates autistic features associated with SHANK3 deficiency

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SH3 and multiple ankyrin repeat domains 3 (*SHANK3*) haploinsufficiency is causative for the neurological features of Phelan-McDermid syndrome (PMDS), including a high risk of autism spectrum disorder (ASD). We used unbiased, quantitative proteomics to identify changes in the phosphoproteome of Shank3-deficient neurons. Down-regulation of protein kinase B (PKB/Akt)–mammalian target of rapamycin complex 1 (mTORC1) signaling resulted from enhanced phosphorylation and activation of serine/threonine protein phosphatase 2A (PP2A) regulatory subunit, B56β, due to increased steady-state levels of its kinase, Cdc2-like kinase 2 (CLK2). Pharmacological and genetic activation of Akt or inhibition of CLK2 relieved synaptic deficits in Shank3-deficient and PMDS patient-derived neurons. CLK2 inhibition also restored normal sociability in a Shank3-deficient mouse model. Our study thereby provides a novel mechanistic and potentially therapeutic understanding of deregulated signaling downstream of Shank3 deficiency.

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Chromosomal aberrations at 22q13 that delete or inactivate one SH3 and multiple ankyrin repeat domains 3 (*SHANK3*) allele are genetic hallmarks of Phelan-McDermid syndrome (PMDS). De novo mutations in *SHANK3* are also associated with nonsyndromic autism spectrum disorder (ASD) and intellectual disability (*1*–*4*). Genetic ablation of *Shank3* in mice yields ASD-like behavioral phenotypes and synaptic dysfunction (*5*–*10*), the latter of which has also been observed in PMDS patient-derived neurons or following Shank3 knockdown in primary rodent neurons (*11*, *12*). Shank3 is a large protein that serves as a scaffold to organize excitatory postsynaptic densities through protein-protein interactions [reviewed in (*13*)]. Reduced expression of Shank3 results in decreased dendritic spine density (*6*, *7*, *11*, *14*), whereas its overexpression enhances spine number or induces spine formation in aspiny neurons (*14*, *15*).

PMDS patients treated with insulin-like growth factor-1 (IGF-1) showed improvements in social and stereotyped behaviors (*16*). IGF-1 also alleviated deficits in Shank3-deficient mice (*17*) and PMDS neurons (*12*). Here, we used phosphoproteomics to investigate the signaling pathways that link Shank3 and IGF-1.

We used rat cortical neurons and reduced Shank3 expression by knockdown. We treated these neurons with brain-derived neurotrophic factor (BDNF) to elicit synaptic plasticity-dependent changes and then analyzed the phosphoproteome (fig. S1). Several phosphopeptides originating from protein substrates in the protein kinase B (PKB or Akt)-mammalian target of rapamycin complex 1 (mTORC1) pathway showed reduced phosphorylation levels in Shank3 knockdown neurons (Fig. 1A, table S1, and database S1). In particular, phosphorylation on GSK3β and ribosomal protein S6 (rpS6) was less than normal on Akt and mTORC1-dependent sites, respectively (Fig. 1A and table S1). Phosphorylation on, B56β (*Ppp2r5b*), a PP2A phosphatase regulatory subunit (*18*), was elevated on a peptide sequence known (*19*) to regulate dephosphorylation of Akt (Fig. 1, A and B). No significant changes were observed in extracellular–signal regulated kinase (ERK) signaling (Mapk1/3) (Fig. 1A and table S1). Consistent with this we also observed reduced pathway activity in synaptosomal fractions from a newly produced Shank3-deficient mouse model (table S2, database S2, and fig. S9). In a second method of proteomic analysis, we enriched less mTOR from Shank3 knockdown neurons on the basis of affinity for pan-kinase inhibitor conjugated beads (Kinomatrix beads) (*20*) (fig. S1 and database S3). The Akt-mTORC1 signaling pathway is linked to ASD (*21*, *22*) and is regulated by IGF-1, which is currently being explored as a therapeutic for PMDS (*12*, *16*, *17*). We therefore explored whether Shank3 deficiency in PMDS leads to re-

duced Akt-mTORC1 signaling by way of enhanced PP2A-mediated inactivation of Akt (fig. S1).

Shank3 knockdown in primary neurons caused a two-fold reduction of Akt phosphorylation at T308 in basal conditions or following BDNF stimulation (Fig. 1C) but had less of an effect at S473. Coincident phosphorylation of both Akt sites (T308 and S473) is required for full Akt activation. Phosphorylation of rpS6 was similarly reduced (Fig. 1C). Attenuated Akt T308 phosphorylation was observed with two additional Shank3 shRNAs and was rescued by co-expression of nontargeted Shank3 (fig. S2). mTOR steady-state levels were unchanged. This indicated that reduced mTOR recovery in Kinomatrix capture, which enriches activated kinases through affinity for a promiscuous ATP-competitive inhibitor resin, was due to impaired kinase activity following Shank3 knockdown. BDNF and its receptor TrkB also activate ERK and Phospholipase C_{γ} (PLC_{γ}) pathways (fig. S2), however, no impairment was observed in either of these pathways (Fig. 1C and fig. S2). We observed no change in Akt phosphorylation after knockdown of another postsynaptic density protein, PSD95, thus indicating that impaired Akt signaling is specific to Shank3 deficiency rather than generic reduction of synaptic proteins (fig. S2). Phosphorylation of the BDNF receptor, TrkB, and expression of the lipid phosphatase PTEN were also unchanged (Fig. 1C), suggesting that attenuated Akt-mTORC1 signaling occurs directly through deregulated Akt phosphorylation. These findings were corroborated in human iPSC-derived neurons from two PMDS patients who harbor deletions within the *SHANK3* locus (fig. S3). As with Shank3 knockdown, Akt T308 phosphorylation was reduced in these SHANK3-deficient neurons (Fig. 1, D and E).

Next, we examined the possibility that Akt-signaling is attenuated by elevated PP2A dephosphorylation activity. B56β, a regulatory subunit of PP2A, is expressed in the brain and defines PP2A specificity for Akt T308 dephosphorylation (fig. S1) (*18*, *23*). Phosphorylated B56β recruits PP2A catalytic (PP2Ac) and structural subunits to Akt for heterotrimeric phosphatase holoenzyme assembly and T308 dephosphorylation (*19*). In Shank3 knockdown neurons, association of Akt with PP2Ac was two-fold above normal (Fig. 2A). Thus, excess of B56β phosphorylation enhances recruitment of PP2A to Akt and reduces phosphorylation of Akt within the catalytic domain at T308 thereby attenuating activity. Addition of the PP2A inhibitor Okadaic acid (OA), alone or together with BDNF, increased Akt T308 phosphorylation in Shank3 knockdown neurons by two-fold, relative to control (Fig. 2B). Expression of a B56β variant lacking the regulatory phosphorylation sites (B56β 6A) (*19*) restored Akt phosphorylation in Shank3 knockdown neurons to control levels (Fig. 2C; compare lanes 1, 2 with 11, 12). Thus, Shank3 deficiency in neurons engenders enhanced activity of B56β/PP2A leading in turn to excessively dephosphorylated Akt and deficiency in Akt activity.

B56β is phosphorylated by CLK2 (*19*) on serine residues in the regulatory peptide sequence. Shank3 knockdown neurons exhibited a two-fold increase in CLK2 protein levels (Fig. 2D), consistent with excess phosphorylation on B56β (Fig. 1, A and B). In hepatocytes, CLK2 is up-regulated by insulin induction to homeostatically regulate Akt via B56β/PP2A-mediated dephosphorylation (*19*, *24*). BDNF treatment induced the accumulation of CLK2 in control neurons but not in Shank3 knockdown neurons (fig. S4). This suggests that the regulated expression of CLK2 by ubiquitination (*24*) is lost in the absence of Shank3. Proteasome inhibition caused a rapid increase of CLK2 in control neurons but not in neurons deficient in Shank3 (fig. S4). Immunoprecipitation of Myc-CLK2 showed less CLK2 ubiquitination in Shank3 knockdown neurons (Fig. 2E). Shank3 knockdown neurons exhibited no change in CLK2 mRNA levels (fig. S5). Treatment with an ATP-competitive CLK2 inhibitor, TG003 (fig. S4), restored Akt T308 phosphorylation in Shank3 knockdown neurons to control levels (Fig. 2F). Thus, increased steady-state levels of CLK2 in Shank3-deficient neurons occurs via impaired ubiquitination and causes excess Akt inactivation through elevated B56β/PP2A activity (fig. S4).

We anticipated that Akt activation, either directly or via CLK2 inhibition, would reverse neuronal impairments associated with Shank3 deficiency. Shank3 knockdown in organotypic brain slice cultures decreased dendritic spine density and was rescued by Shank3 re-expression (fig. S6) (*4*, *6*, *11*). We then exposed slices to the small molecule Akt activator SC79 (fig. S7) (*25*) or to TG003. We observed that both treatments restored spine density in Shank3 knockdown neurons to control levels (Fig. 3, A and B). This was dependent on Akt activity as it was blocked by co-incubation with a validated Akt inhibitor (Fig. 3B and fig. S7). Reduction of CLK2 in Shank3 knockdown neurons had a similar restorative effect on spine density to that of TG003 treatment (fig. S6). We confirmed that Akt inhibition in wild type neurons is sufficient to reduce spine density and thereby phenocopy the effect of Shank3 deficiency on reducing spine density via downstream Akt attenuation (fig. S6). Shank3 knockdown in organotypic slices reduced miniature excitatory postsynaptic current (mEPSC) frequency that was restored to control levels with SC79 treatment (Fig. 3C). IPSC-derived neurons from two unrelated PMDS patients exhibited reduced frequency of spontaneous EPSCs (sEPSCs) (Fig. 3D and fig. S8). Treatment with SC79 or TG003 again rescued these deficits, in an Akt-dependent manner (Fig. 3D, highlighted groups). Thus, Akt activation alone or via CLK2 inhibition is sufficient to restore synaptic impairments occurring from Shank3 deficiency.

To determine whether our findings extend to ASD-like behaviors, we generated a Shank3-deficient mouse model by ablation of *Shank3* exon 21 (fig. S9) as previously described (*8*, *10*). The major Shank3 isoforms are absent in our homozygous mice (Shank3$^{\Delta C/\Delta C}$), while faster-migrating, truncated fragments were detected (fig. S9). Shank3 $^{\Delta C/\Delta C}$ neurons displayed excess CLK2 expression (fig. S9). Neither heterozygous (Shank3 $^{+/\Delta C}$) nor Shank3 $^{\Delta C/\Delta C}$ mice exhibited anxious behavior or locomotor skill impairments (fig. S10). Both Shank3 $^{+/\Delta C}$ and Shank3 $^{\Delta C/\Delta C}$ mice displayed avoidance behavior, assessed by marble burying, that was refractory to treatment with TG003 (fig. S10). In contrast, we observed that only Shank3 $^{\Delta C/\Delta C}$ mice exhibited excess self-grooming, a trait reflecting repetitive behaviors seen in ASD (fig. S10). Treatment of these mice with TG003 significantly decreased self-grooming although not to wild type frequency (fig. S10). Mice were then tested in a social motivation paradigm (fig. S10). Wild type and Shank3$^{+/\Delta C}$ mice displayed a significant preference for social investigation, whereas Shank3 $^{\Delta C/\Delta C}$ mice did not (Fig. 4, A and B, and fig. S10). In contrast, Shank3 $^{\Delta C/\Delta C}$ mice treated with TG003 recovered normal preference for social interaction (Fig. 4, A and B, and fig. S10). We observed that this effect persisted three days after treatment when a new cohort of animals was tested (fig. S10). The recovery of normal social behavior correlated with restored Akt phosphorylation in synaptosomal fractions (fig. S9). No significant preference was observed between groups when a second intruder was introduced (Fig. 4B). Thus, CLK2 inhibition rescues deficits in social behavior caused by Shank3 deficiency.

Mutations in Akt-mTORC1 pathway regulators PTEN and TSC1/2 yield clinical syndromes that include features of ASD by means of exaggerated Akt-mTORC1 activity (*21*, *22*). However, attenuated Akt-mTORC1 activity has also been associated with Angelman syndrome, another monogenic form of ASD (*26*, *27*). This suggests a bidirectional regulation of the Akt-mTORC1 pathway associated with different ASD genes. Here, we show that Shank3 deficiency leads to enhanced CLK2 expression and attenuated Akt-mTORC1 activity. Cellular and behavioral impairments attributed to Shank3 loss of function have been corrected with IGF-1 treatment (*12*, *16*, *17*). We found that IGF-1 treatment restored normal dendritic spine density to Shank3 knockdown neurons, in an Akt-dependent manner (fig. S11). We suggest that IGF-1 restores balance in signaling pathways by boosting Akt phosphorylation to counteract elevated dephosphorylation by PP2A (fig. S12). Thus, we propose that direct Akt-reactivation or CLK2 inhibition may be therapeutic targets for intervention in patients with PMDS.

REFERENCES AND NOTES

1. K. Phelan, H. E. McDermid, The 22q13.3 deletion syndrome (Phelan-McDermid Syndrome). *Molecular Syndromology* **2**, 186–201 (2012). Medline
2. C. Redin, B. Gérard, J. Lauer, Y. Herenger, J. Muller, A. Quartier, A. Masurel-Paulet, M. Willems, G. Lesca, S. El-Chehadeh, S. Le Gras, S. Vicaire, M. Philipps, M. Dumas, V. Geoffroy, C. Feger, N. Haumesser, Y. Alembik, M. Barth, D. Bonneau, E. Colin, H. Dollfus, B. Doray, M. A. Delrue, V. Drouin-Garraud, E. Flori, M. Fradin, C. Francannet, A. Goldenberg, S. Lumbroso, M. Mathieu-Dramard, D. Martin-Coignard, D. Lacombe, G. Morin, A. Polge, S. Sukno, C. Thauvin-Robinet, J. Thevenon, M. Doco-Fenzy, D. Genevieve, P. Sarda, P. Edery, B. Isidor, B. Jost, L. Olivier-Faivre, J. L. Mandel, A. Piton, Efficient strategy for the molecular diagnosis of intellectual disability using targeted high-throughput sequencing. *J. Med. Genet.* **51**, 724–736 (2014). Medline doi:10.1136/jmedgenet-2014-102554
3. J. Gauthier, N. Champagne, R. G. Lafrenière, L. Xiong, D. Spiegelman, E. Brustein, M. Lapointe, H. Peng, M. Côté, A. Noreau, F. F. Hamdan, A. M. Addington, J. L. Rapoport, L. E. Delisi, M. O. Krebs, R. Joober, F. Fathalli, F. Mouaffak, A. P. Haghighi, C. Néri, M. P. Dubé, M. E. Samuels, C. Marineau, E. A. Stone, P. Awadalla, P. A. Barker, S. Carbonetto, P. Drapeau, G. A. Rouleau; S2D Team, De novo mutations in the gene encoding the synaptic scaffolding protein SHANK3 in patients ascertained for schizophrenia. *Proc. Natl. Acad. Sci. U.S.A.* **107**, 7863–7868 (2010). Medline doi:10.1073/pnas.0906232107
4. C. M. Durand, C. Betancur, T. M. Boeckers, J. Bockmann, P. Chaste, F. Fauchereau, G. Nygren, M. Rastam, I. C. Gillberg, H. Anckarsäter, E. Sponheim, H. Goubran-Botros, R. Delorme, N. Chabane, M. C. Mouren-Simeoni, P. de Mas, E. Bieth, B. Rogé, D. Héron, L. Burglen, C. Gillberg, M. Leboyer, T. Bourgeron, Mutations in the gene encoding the synaptic scaffolding protein SHANK3 are associated with autism spectrum disorders. *Nat. Genet.* **39**, 25–27 (2007). Medline doi:10.1038/ng1933
5. O. Bozdagi, T. Sakurai, D. Papapetrou, X. Wang, D. L. Dickstein, N. Takahashi, Y. Kajiwara, M. Yang, A. M. Katz, M. L. Scattoni, M. J. Harris, R. Saxena, J. L. Silverman, J. N. Crawley, Q. Zhou, P. R. Hof, J. D. Buxbaum, Haploinsufficiency of the autism-associated Shank3 gene leads to deficits in synaptic function, social interaction, and social communication. *Molecular Autism* **1**, 15 (2010). Medline doi:10.1186/2040-2392-1-15
6. J. Peça, C. Feliciano, J. T. Ting, W. Wang, M. F. Wells, T. N. Venkatraman, C. D. Lascola, Z. Fu, G. Feng, Shank3 mutant mice display autistic-like behaviours and striatal dysfunction. *Nature* **472**, 437–442 (2011). Medline doi:10.1038/nature09965
7. X. Wang, P. A. McCoy, R. M. Rodriguiz, Y. Pan, H. S. Je, A. C. Roberts, C. J. Kim, J. Berrios, J. S. Colvin, D. Bousquet-Moore, I. Lorenzo, G. Wu, R. J. Weinberg, M. D. Ehlers, B. D. Philpot, A. L. Beaudet, W. C. Wetsel, Y. H. Jiang, Synaptic dysfunction and abnormal behaviors in mice lacking major isoforms of Shank3. *Hum. Mol. Genet.* **20**, 3093–3108 (2011). Medline doi:10.1093/hmg/ddr212
8. M. Kouser, H. E. Speed, C. M. Dewey, J. M. Reimers, A. J. Widman, N. Gupta, S. Liu, T. C. Jaramillo, M. Bangash, B. Xiao, P. F. Worley, C. M. Powell, Loss of predominant Shank3 isoforms results in hippocampus-dependent impairments in behavior and synaptic transmission. *J. Neurosci.* **33**, 18448–18468 (2013). Medline doi:10.1523/JNEUROSCI.3017-13.2013
9. M. Yang, O. Bozdagi, M. L. Scattoni, M. Wöhr, F. I. Roullet, A. M. Katz, D. N. Abrams, D. Kalikhman, H. Simon, L. Woldeyohannes, J. Y. Zhang, M. J. Harris, R. Saxena, J. L. Silverman, J. D. Buxbaum, J. N. Crawley, Reduced excitatory neurotransmission and mild autism-relevant phenotypes in adolescent Shank3 null mutant mice. *J. Neurosci.* **32**, 6525–6541 (2012). Medline doi:10.1523/JNEUROSCI.6107-11.2012
10. L. J. Duffney, P. Zhong, J. Wei, E. Matas, J. Cheng, L. Qin, K. Ma, D. M. Dietz, Y. Kajiwara, J. D. Buxbaum, Z. Yan, Autism-like deficits in Shank3-deficient mice are rescued by targeting actin regulators. *Cell Reports* **11**, 1400–1413 (2015). Medline doi:10.1016/j.celrep.2015.04.064
11. C. Verpelli, E. Dvoretskova, C. Vicidomini, F. Rossi, M. Chiappalone, M. Schoen, B. Di Stefano, R. Mantegazza, V. Broccoli, T. M. Böckers, A. Dityatev, C. Sala, Importance of Shank3 protein in regulating metabotropic glutamate receptor 5 (mGluR5) expression and signaling at synapses. *J. Biol. Chem.* **286**, 34839–34850 (2011). Medline doi:10.1074/jbc.M111.258384
12. A. Shcheglovitov, O. Shcheglovitova, M. Yazawa, T. Portmann, R. Shu, V. Sebastiano, A. Krawisz, W. Froehlich, J. A. Bernstein, J. F. Hallmayer, R. E.

Dolmetsch, SHANK3 and IGF1 restore synaptic deficits in neurons from 22q13 deletion syndrome patients. *Nature* **503**, 267–271 (2013). Medline
13. Y. H. Jiang, M. D. Ehlers, Modeling autism by SHANK gene mutations in mice. *Neuron* **78**, 8–27 (2013). Medline doi:10.1016/j.neuron.2013.03.016
14. G. Roussignol, F. Ango, S. Romorini, J. C. Tu, C. Sala, P. F. Worley, J. Bockaert, L. Fagni, Shank expression is sufficient to induce functional dendritic spine synapses in aspiny neurons. *J. Neurosci.* **25**, 3560–3570 (2005). Medline doi:10.1523/JNEUROSCI.4354-04.2005
15. K. Han, J. L. Holder Jr., C. P. Schaaf, H. Lu, H. Chen, H. Kang, J. Tang, Z. Wu, S. Hao, S. W. Cheung, P. Yu, H. Sun, A. M. Breman, A. Patel, H. C. Lu, H. Y. Zoghbi, SHANK3 overexpression causes manic-like behaviour with unique pharmacogenetic properties. *Nature* **503**, 72–77 (2013). Medline doi:10.1038/nature12630
16. A. Kolevzon, L. Bush, A. T. Wang, D. Halpern, Y. Frank, D. Grodberg, R. Rapaport, T. Tavassoli, W. Chaplin, L. Soorya, J. D. Buxbaum, A pilot controlled trial of insulin-like growth factor-1 in children with Phelan-McDermid syndrome. *Molecular Autism* **5**, 54 (2014). Medline doi:10.1186/2040-2392-5-54
17. O. Bozdagi, T. Tavassoli, J. D. Buxbaum, Insulin-like growth factor-1 rescues synaptic and motor deficits in a mouse model of autism and developmental delay. *Molecular Autism* **4**, 9 (2013). Medline doi:10.1186/2040-2392-4-9
18. S. Padmanabhan, A. Mukhopadhyay, S. D. Narasimhan, G. Tesz, M. P. Czech, H. A. Tissenbaum, A PP2A regulatory subunit regulates *C. elegans* insulin/IGF-1 signaling by modulating AKT-1 phosphorylation. *Cell* **136**, 939–951 (2009). Medline doi:10.1016/j.cell.2009.01.025
19. J. T. Rodgers, R. O. Vogel, P. Puigserver, Clk2 and B56β mediate insulin-regulated assembly of the PP2A phosphatase holoenzyme complex on Akt. *Mol. Cell* **41**, 471–479 (2011). Medline doi:10.1016/j.molcel.2011.02.007
20. C. M. Gower, J. R. Thomas, E. Harrington, J. Murphy, M. E. Chang, I. Cornella-Taracido, R. K. Jain, M. Schirle, D. J. Maly, Conversion of a single polypharmacological agent into selective bivalent inhibitors of intracellular kinase activity. *ACS Chem. Biol.* **11**, 121–131 (2016). Medline
21. A. Goffin, L. H. Hoefsloot, E. Bosgoed, A. Swillen, J. P. Fryns, PTEN mutation in a family with Cowden syndrome and autism. *Am. J. Med. Genet.* **105**, 521–524 (2001). Medline doi:10.1002/ajmg.1477
22. S. L. Smalley, P. E. Tanguay, M. Smith, G. Gutierrez, Autism and tuberous sclerosis. *J. Autism Dev. Disord.* **22**, 339–355 (1992). Medline doi:10.1007/BF01048239
23. B. McCright, D. M. Virshup, Identification of a new family of protein phosphatase 2A regulatory subunits. *J. Biol. Chem.* **270**, 26123–26128 (1995). Medline doi:10.1074/jbc.270.44.26123
24. J. T. Rodgers, W. Haas, S. P. Gygi, P. Puigserver, Cdc2-like kinase 2 is an insulin-regulated suppressor of hepatic gluconeogenesis. *Cell Metab.* **11**, 23–34 (2010). Medline doi:10.1016/j.cmet.2009.11.006
25. H. Jo, S. Mondal, D. Tan, E. Nagata, S. Takizawa, A. K. Sharma, Q. Hou, K. Shanmugasundaram, A. Prasad, J. K. Tung, A. O. Tejeda, H. Man, A. C. Rigby, H. R. Luo, Small molecule-induced cytosolic activation of protein kinase Akt rescues ischemia-elicited neuronal death. *Proc. Natl. Acad. Sci. U.S.A.* **109**, 10581–10586 (2012). Medline doi:10.1073/pnas.1202810109
26. S. Ricciardi, E. M. Boggio, S. Grosso, G. Lonetti, G. Forlani, G. Stefanelli, E. Calcagno, N. Morello, N. Landsberger, S. Biffo, T. Pizzorusso, M. Giustetto, V. Broccoli, Reduced AKT/mTOR signaling and protein synthesis dysregulation in a Rett syndrome animal model. *Hum. Mol. Genet.* **20**, 1182–1196 (2011). Medline doi:10.1093/hmg/ddq563
27. C. Cao, M. S. Rioult-Pedotti, P. Migani, C. J. Yu, R. Tiwari, K. Parang, M. R. Spaller, D. J. Goebel, J. Marshall, Impairment of TrkB-PSD-95 signaling in Angelman syndrome. *PLOS Biol.* **11**, e1001478 (2013). Medline doi:10.1371/journal.pbio.1001478

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SUPPLEMENTARY MATERIALS

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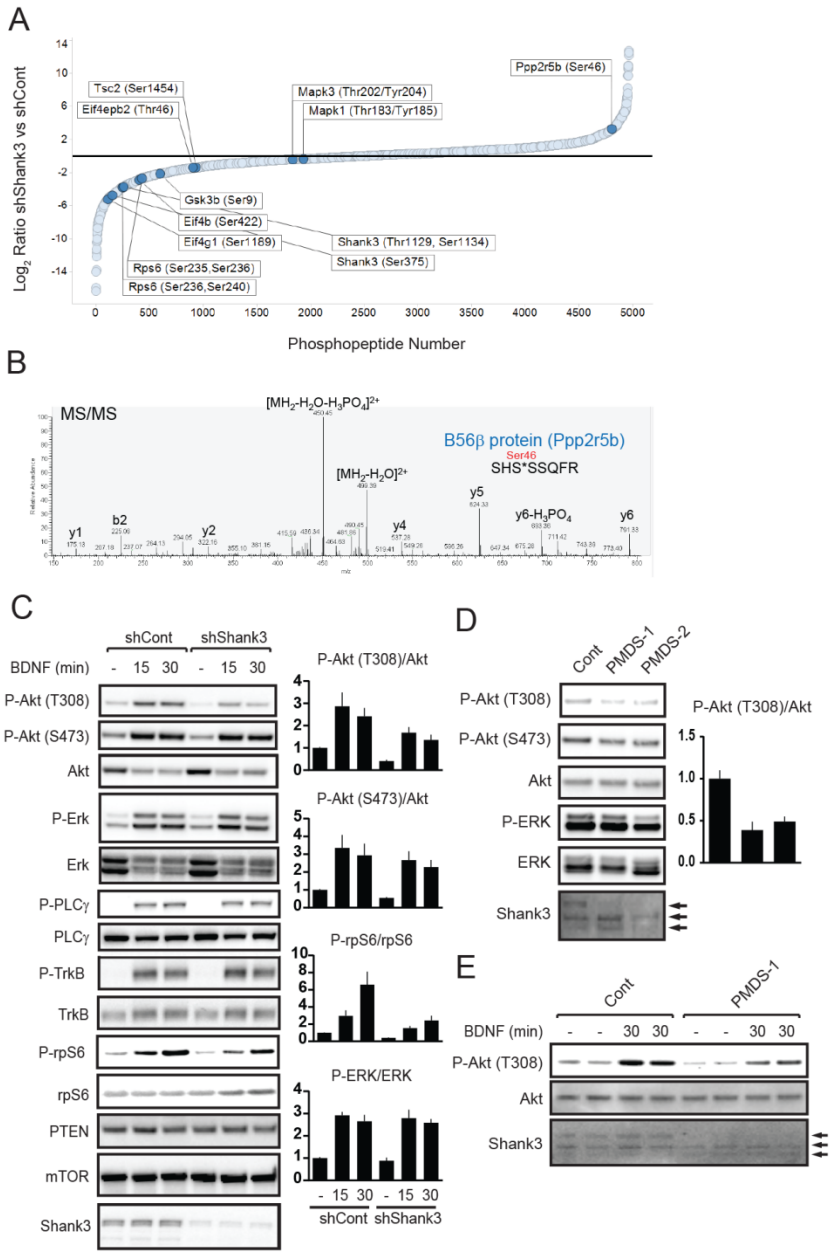

Fig. 1. Shank3 deficiency impairs Akt-mTORC1 signaling. (**A**) Relative quantitation of phosphopeptide abundance between Shank3 knockdown and control neuron samples. Down-regulation of phosphopeptides from several proteins targeted by Akt-mTORC1 signaling were detected, as was up-regulation of a phosphopeptide from PP2A regulatory subunit, B56β. Nonsignificant changes of Mapk1/Mapk3 are shown in the center. (**B**) Representative tandem MS/MS spectrum of B56β phosphopeptide SHS*SSQFR identified with increased abundance (Log_2FC = 2.489) in Shank3 knockdown neurons. (**C**) Western blotting validation of reduced Akt-mTORC1 signaling in Shank3 knockdown neurons. Neurons were treated with BDNF (50 ng/ml) and cell lysates were prepared in RIPA buffer. Plots are means ± SEM. (**D**) Impaired Akt activity in PMDS iPSC-derived patient neurons. Cell lysates were prepared as in (C) at 8 weeks in vitro. Plots are means ± SEM. (**E**) Impaired Akt activation following BDNF stimulation in PMDS iPSC-derived patient neurons. Cell lysates were prepared as in (C) at 8 weeks in vitro.

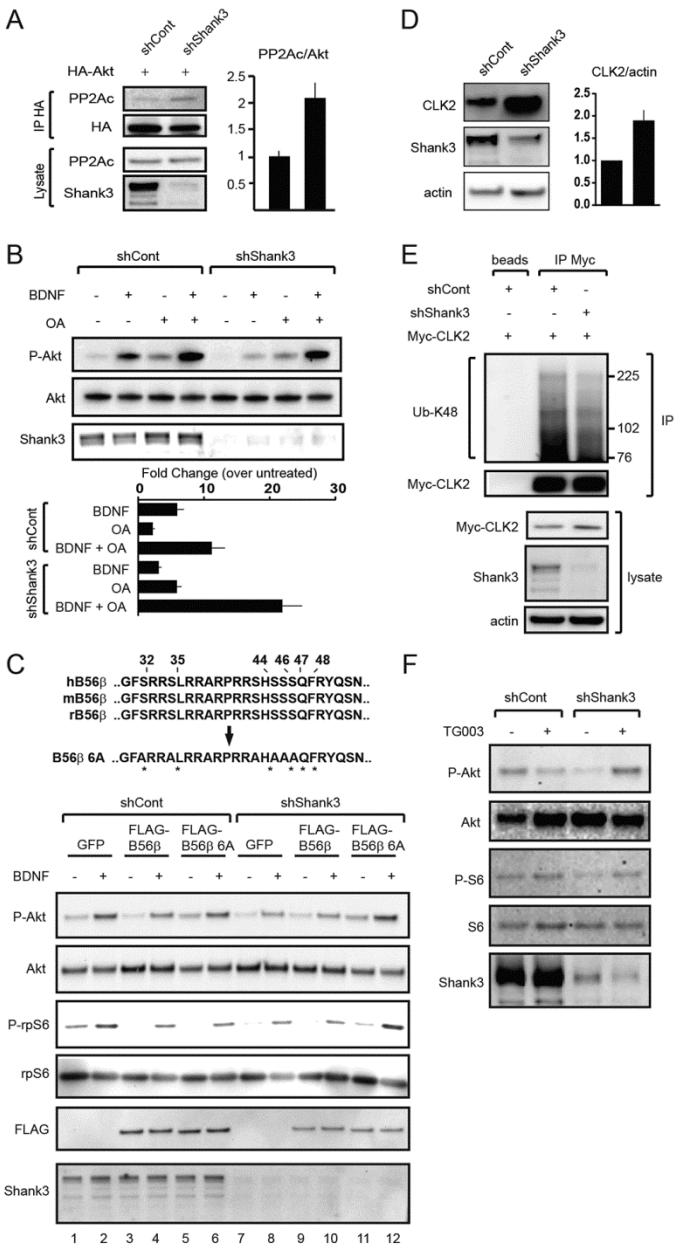

Fig. 2. Elevated B56β/PP2A activity and CLK2 expression in Shank3-deficient neurons. (**A**) Association of PP2A catalytic subunit (PP2Ac) with Akt is enhanced in Shank3 knockdown neurons. Primary neurons co-transduced with shRNA and HA-Akt lentiviruses were lysed and immunoprecipitated with HA antibody. (**B**) PP2A-inhibition restores Akt activity in Shank3 knockdown neurons. Neurons were treated with 100 nM Okadaic acid 15 min prior to and during BDNF (50 ng/ml) incubation, followed by lysis and Western blotting. Data are means ± SEM. (**C**) Overexpression of a phosphorylation-defective variant of B56β restores Akt activity in Shank3 knockdown neurons. Flag-tagged B56β, or a variant lacking phospho-serines on the indicated sites (B56β 6A) (*19*), were co-transduction with shRNA viruses. Neurons were treated with BDNF followed by and Western blotting. (**D**) Increased steady-state levels of CLK2 in Shank3 knockdown neurons. Neurons were transduced with lentiviruses followed by Western blotting. Data are means ± SEM. (**E**) Attenuated CLK2 ubiquitination in Shank3 knockdown neurons. Neurons were co-transduced with shRNA and Myc-CLK2 lentiviruses. Cell lysates were prepared in IP buffer and immunoprecipitated with anti-Myc antibody followed by Western blotting for the proteasome-targeting Ubiquitin K48-linkage. (**F**) CLK2 inhibition restores Akt activity in Shank3-deficient neurons. Neurons were treated with 20 μM TG003 for 60 min prior to Western blotting.

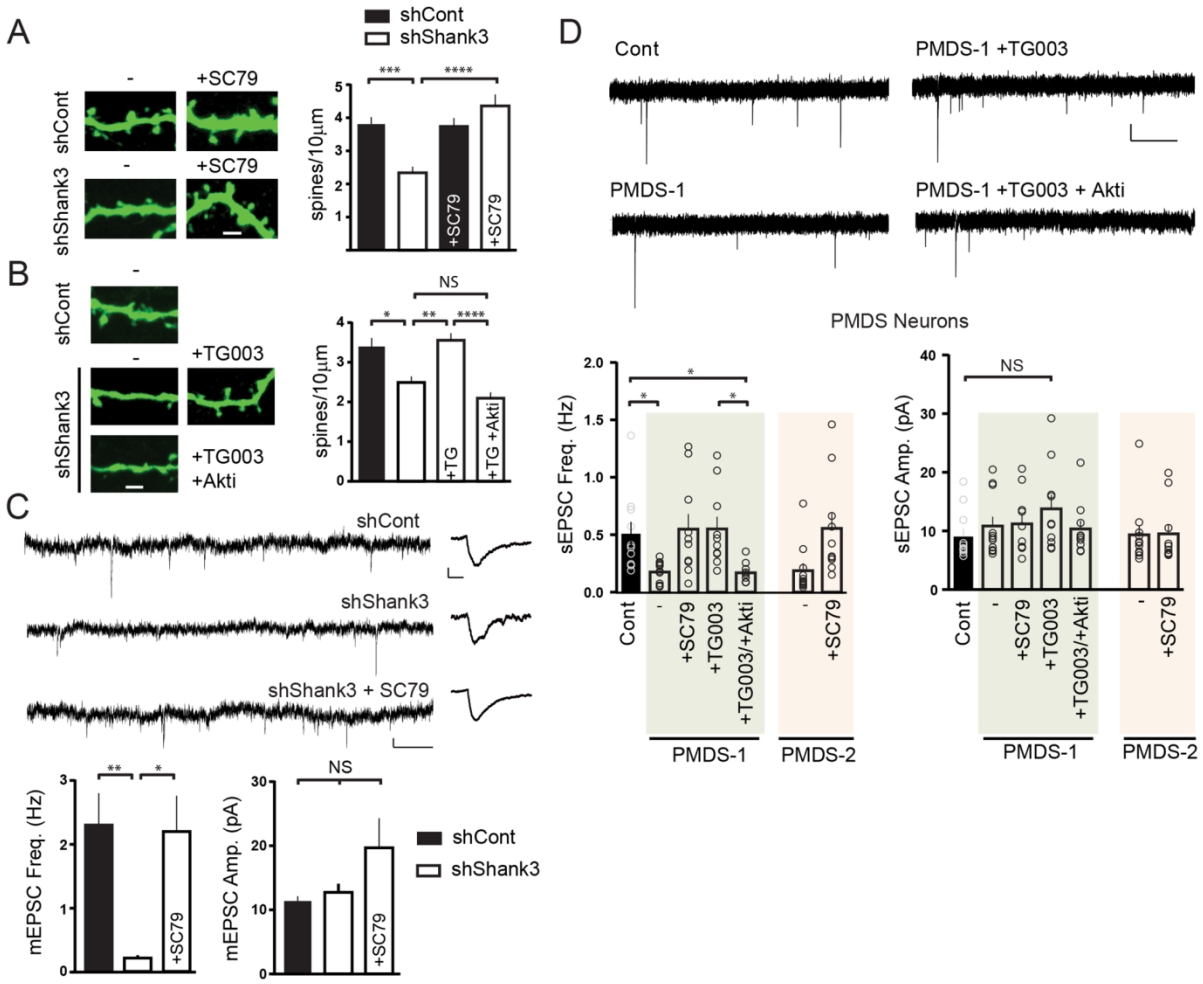

Fig. 3. Activation of Akt or inhibition of CLK2 corrects Shank3-associated deficits. (**A**) SC79 rescues spine density in Shank3 knockdown neurons. Neurons were treated on DIV 14 for 24 hours with 4 µg/ml SC79 prior to immunostaining for GFP. Plots indicate group means ± SEM (one-way ANOVA, $P < 0.0001$, Dunn's multiple comparisons test). (**B**) CLK2 inhibition restores normal spine density in Shank3 knockdown neurons. Organotypic slices were treated for 24 hours with 20 µM TG003 and 10 µM Akt inhibitor (Akti), as indicated. (one-way ANOVA, $P < 0.0001$, Dunn's multiple comparisons test). Scale bar is 2 µm. (**C**) SC79 rescues impaired synaptic transmission frequency in Shank3 knockdown neurons. (Upper left) Representative traces of mEPSCs recorded from CA1 neurons in organotypic slice cultures prepared and treated as in (A) and fig. S6 (scale bar: 5 pA, 1 s). (Upper right) Average traces of the mEPSCs amplitude for the cells shown (scale bar: 1.6 pA, 5 ms). (Lower) Plots of recorded mEPSCs. Data are means ± SEM (one-way ANOVA, $P < 0.005$, Tukey's multiple comparisons test). (**D**) SC79 and TG003 correct synaptic transmission impairments in PMDS neurons. (Upper) Representative traces of recorded sEPSCs (scale bar: 20 pA, 5 s). (Lower) Plots of 24 hours treatments on sEPSC frequency and amplitude of PMDS patient-derived neurons at eight weeks in vitro. Data are means ± SEM (one-way ANOVA, $P < 0.0001$, Dunn's multiple comparisons test).

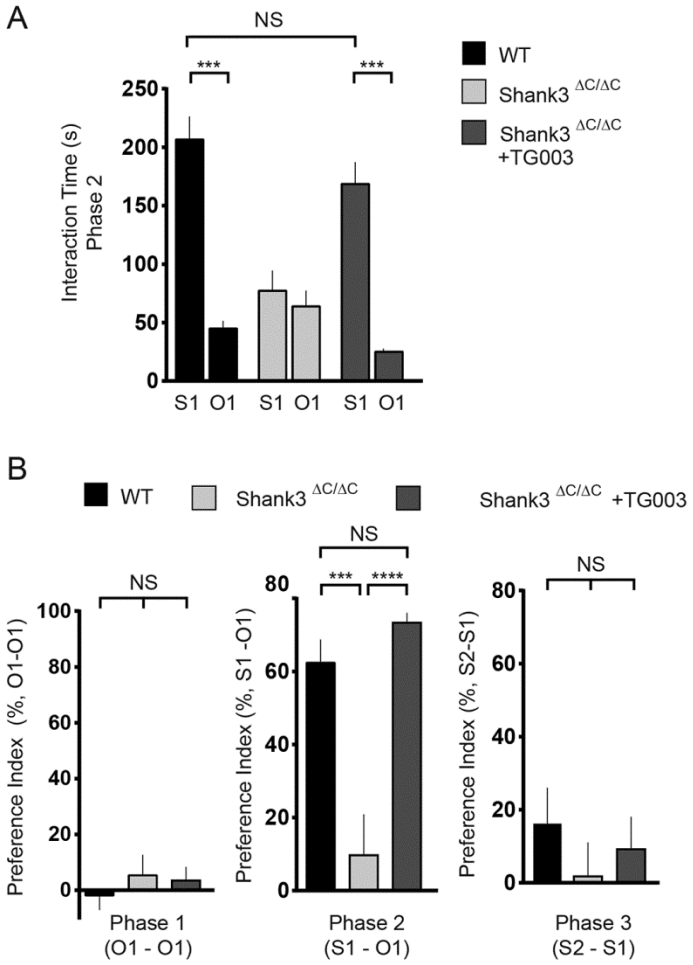

Fig. 4. CLK2 inhibition corrects impaired social motivation in Shank3$^{\Delta C/\Delta C}$ mice. (**A**) Shank3$^{\Delta C/\Delta C}$ mice display impaired motivation for social interaction that is corrected by treatment with CLK2-inhibitor, TG003. Interaction times with the intruder mouse (S1) or the object (O1) are plotted for phase 2. Data are means ± SEM with paired *t* tests (WT $P < 0.0005$; Shank3$^{\Delta C/\Delta C}$ +TG003 $P < 0.0005$) comparing S1 to O1 investigation times within each group. Comparison of social interaction times across groups was by one-way ANOVA with Tukey's multiple comparisons test ($P < 0.0005$ for differences amongst group means). (**B**) Preference index for S1 versus O1 of interaction times (see Methods) was calculated for each test phase. Data are means ± SEM (one-way ANOVA, $P < 0.0001$, Tukey's multiple comparisons test).